

PowerShell Cmdlets

Cmdlets are basic, lightweight, built-in functions in PowerShell that perform specific tasks.

They follow 'verb-noun' structure where verb is represented as their action ('Get', 'Set', 'New', etc.) and noun is the object of the action ('Process', 'Service', etc.)

Examples include Get-Process, Start-Service, and New-Item. These cmdlets are essential for automating tasks and managing the system.

Basic Verbs used are as follows:

Get: Used to Retrieve Information

Set: Used to change some value

New: Used to create Instance

Remove: Used to delete something

Key Cmdlets and their uses are listed below as follows:

Get-Command: Lists all available PowerShell commands, including cmdlets, functions helping users to discover and understand the capabilities of PowerShell.

Get-Process: Retrieves information about running processes

Start-Service: Starts a specified service.

Stop-Service: Stops a specified service.

New-Item: Creates new files, directories, or registry keys.

Get-Item: Retrieves information about existing files, directories, or registry keys.

Set-Item: Modifies properties of existing files, directories, or registry keys.

Write-Output: Displays output to the console.

Out-File: Writes output to a file.

Read-Host: Reads input from the user.

Get-PSDrive: Displays information about existing PowerShell drives.

Pipelines in PowerShell are used as follows:

Get-Process | Sort-Object CPU -Descending | Select-Object -First 5 | Format-Table Name, CPU

Filtering with Where-Object:

The Where-Object cmdlet filters objects based on specified criteria, It also uses the PowerShell **pipeline for sequential execution of commands.**

Eg: **Get-Process | Where-Object {\$_.CPU -gt 1}**

Multiple conditions can be combined using the logical operators

Eg : **Get-Process | Where-Object {\$_.CPU -gt 1 -and \$_.WorkingSet -gt 10MB}**

Input-Output in PowerShell are as follows:

1. **Read-Host:** This cmdlet prompts the user for input and stores it as a string.

Eg: \$name = Read-Host "Please enter your name"

2. **Get-Content:** Reads the content of a file.

Eg: \$content = Get-Content -Path "C:\example.log"

3. **Write-Host:** Displays output directly to the console.

Eg: Write-Host "This is a message."

4. **Out-File:** Writes output to a file.

Eg: Get-Process | Out-File -FilePath "processes.txt"

Formatting in PowerShell are as follows:

Format-Table: Displays output in a table format.

Eg: Get-Process | Format-Table -Property Name, CPU, StartTime

Format-List: Displays output as a list of properties.

Eg: Get-Service | Format-List -Property Name, Status, DisplayName

Format-Wide: Displays output in a wide format, showing only one property per line.

Eg: Get-ChildItem | Format-Wide -Column 3